1ページに画像1枚を貼り付けてください。

3

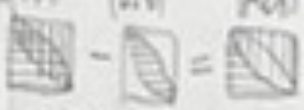
[P1,5]

Xの部分を|X|とすると

$|A\cup B| = |A| + |B| - |A\cap B|$

$|A\cup B| = 1$ で $|A| = |B| = \frac{n}{4}$

$|A|+|B|$ $|A\cap B|$ $|A\cup B|$



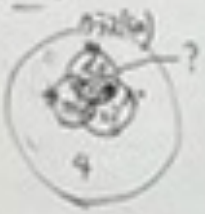
①より

$1 = |A\cup B| = \frac{n}{4} + \frac{n}{4} - |A\cap B|$

よって

$|A\cap B| = \frac{n}{2} - 1$

[P2,1]



$|A\cup B| = |A| + |B| - |A\cap B|$ [P1,11]

$= 33 + 25 - 8$

$= 58$

[P1,9]

(1) $2^4 = P(A)$

$= \{\phi, \{\alpha\}, \{\beta\}, \{\alpha, \beta\}\}$

$P(C) = \{\phi, \{\alpha\}, \{\beta\}, \{\gamma\}, \{\delta\}, \{\alpha, \beta\}, \{\alpha, \gamma\}, \{\alpha, \delta\}, \{\beta, \gamma\}, \{\beta, \delta\}, \{\gamma, \delta\}, \{\alpha, \beta, \gamma, \delta\}\}$

(2) $A\times C = \{(\alpha, \alpha), (\alpha, \beta), (\alpha, \gamma), (\alpha, \delta), (1, \alpha), (1, \beta), (1, 2), (1, \delta)\}$

$B\times A = \{(a, \alpha), (a, 1), (b, \alpha), (b, 1), (c, \alpha), (c, 1)\}$

$B^2 = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$

$A^3 = \{(\alpha, \alpha, \alpha), (\alpha, \alpha, 1), (\alpha, 1, \alpha), (\alpha, 1, 1), (1, \alpha, \alpha), (1, \alpha, 1), (1, 1, \alpha), (1, 1, 1)\}$

[P1,10]

$P(B) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

$C = \{\phi, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

$P(A\cap B) = P(A) \cap P(B) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

[P1,12]

$50 = 30 + 35 - |A\cap B|$

$|A\cap B| = 20$

[P1,13]

$100 = 20 + 35 + 40 - 25 - 20 + 10 - |A\cap B|$

$-|A\cap B| = 100 - 20 - 35 - 40 + 25 + 20 - 10$

$= 125 - 105 = 20$

$|A\cap B| = 20$

$H = A$, $T = B$, $K = C$

$|A| = 10$, $|B| = 25$, $|C| = 33$

$|A\cap B| = 15$, $|A\cap C| = 12$, $|B\cap C| = 10$

$\therefore 56 = 10 + 25 + 33 - 15 - 12 - 10 + |A\cap B\cap C|$

$\therefore |A\cap B\cap C| = 5$

[P1,F]

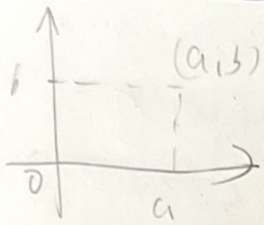
$|A\cup B|$?

$100 \div 3 = 33 = |A|$

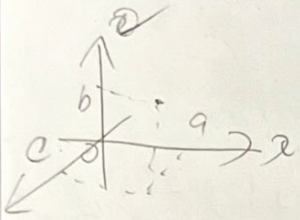
$100 \div 4 = 25 = |B|$

$A\cap B = 12$, $|A\cup B| = 100 \div 12 = 8$

(数学 解答 7
2025.11.5 22059 [illegible]



平面



空間

Q1.1.1 CNFをP = (P1∨P2)∧(¬P1∨¬P2)∧(P1∨¬P3)
∧(P2∨P3)

P1 = T, P2 = F, P3 = T で充足可能

定理1.1

x∈A のとき, A⊂B∧B⊂C なら x∈B∧
それぞれの x∈A は x∈C

A⊂B∧B⊂C ⇒ A⊂C

n個の命題 P1 P2 … Pn の真偽値が与えられて
それぞれ1,2…n番目で真偽が分かっているとき

Q1.1.2 ¬(P∨Q) ≡ (¬P∧Q)

⇔ (¬P∧¬Q) ∨ (¬P∧Q)

⇔ ¬P(¬Q∧Q) ⇔ ¬P・1 ⇔ ¬P

ex 1.1.2

CNFをP = (P1∨P2∨P3)∧(¬P1∨¬P2∨¬P3)∧(P1∨¬P3)

C = {P} とする

Q1.1.3

Q = ¬P, C = φ

a) P∨(P⇒Q) ⇔ T

~~左辺~~ ⇔ P∨(¬P∨Q)

⇔ T∨Q ⇔ T

b) P∧

1 より P2 = T なし

(P1∨T∨P3)∧(¬P1∨T∨¬P3)∧(P1∨¬P3)∧T

= (¬P1∨¬P3)∧(P1∨¬P3)

II より P3 = F のとき Q' = (¬P1∨T)∧(P1∨T) = T

値を出かして終了

$\boxed{2}_2$

$\boxed{2.1}$ $A \subset B \cap B \subset C \Rightarrow A \subset C$

"$x \in A$ のとき $A \subset B \cap B \subset C$ であれば

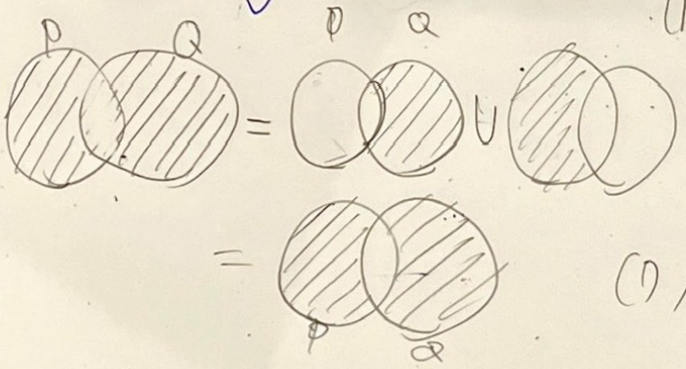
$x \in B \wedge x \in C$ だから $x \in A \Rightarrow x \in C$

このことから成り立つ

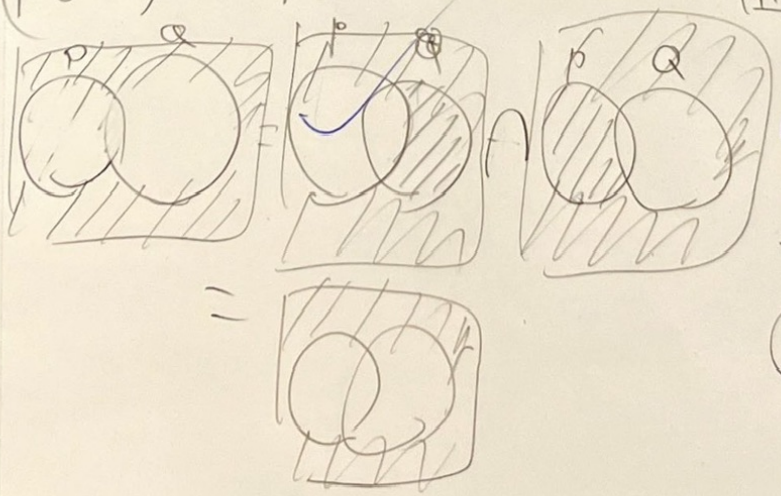
$\boxed{2.2}$ $\cap, \cup, -$ $\{2, 4, 8\}$

$(A \cap B) - D - C)$

$\boxed{2.3}$ $(P \cap Q)^c = P^c \cup Q^c$



$(P \cup Q)^c = P^c \cap Q^c$



$\boxed{2.4}$

a) $X - Y = \{o, f, t, w, a, r, c, i, n\}$ $\{e, s, w\}$

b) $Y \cap Z = \{i, n\}$

c) $Z - (X \cup Y) = \{c\}$

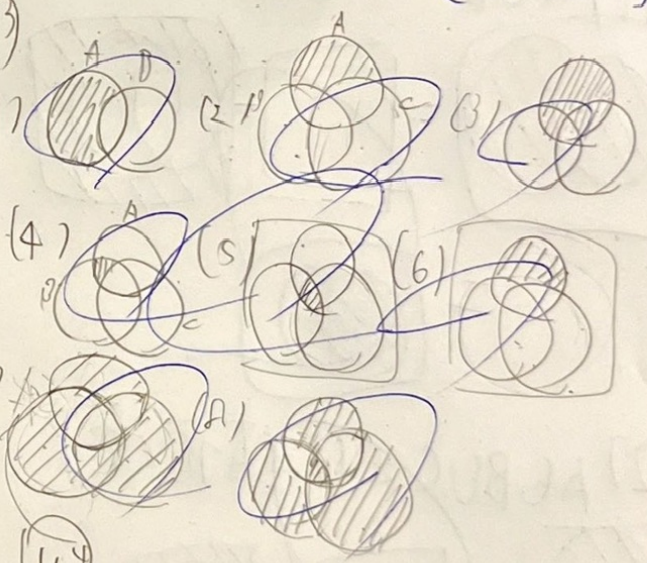
d) $X \cap Y \cap Z = \cancel{\{\emptyset\}}\ \emptyset$

e) $X^c \cap Y \cap Z^c = \{m\}$

f) $X^c \cap (Y \cup Z) = \{s, o, f, t, a, r, e\}$ $\{c, i, m, n\}$

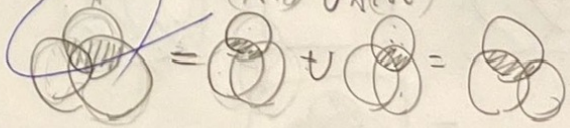
$\boxed{3}$

$\boxed{1.3}$

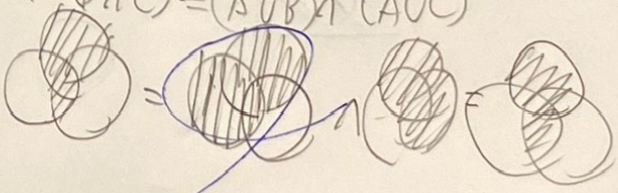


$\boxed{1.4}$

$A \cap (B \cup C) = (A \cap B) \cup A \cap C)$

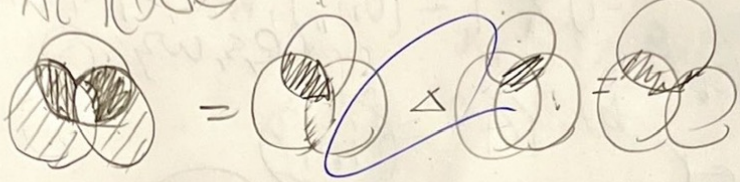


$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

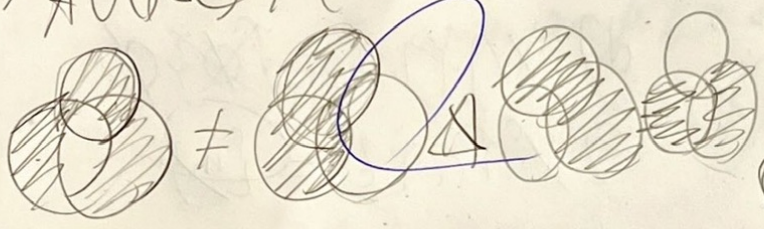


1.5

(1) $A\cap(B\triangle C)=(A\cap B)\triangle(A\cap C)$

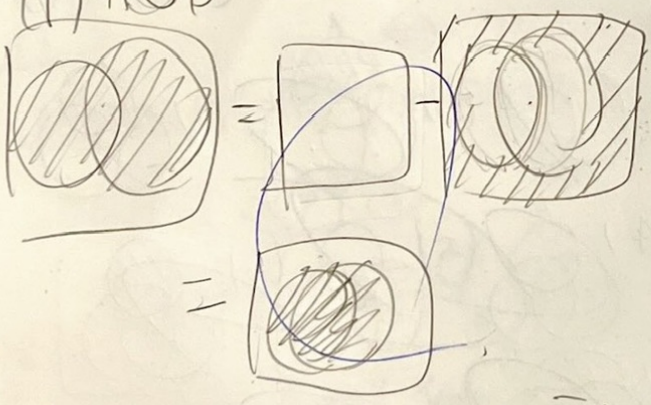


(2) $A\cup(B\triangle C)\neq(A\cup B)\cap(A\cup C)$

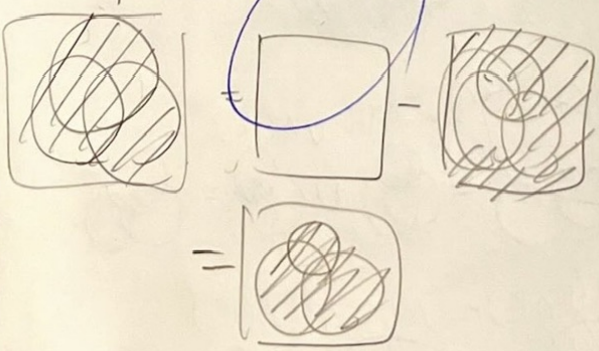


1.6

(1) $A\cup B=U-(\overline{A}\cap\overline{B})$

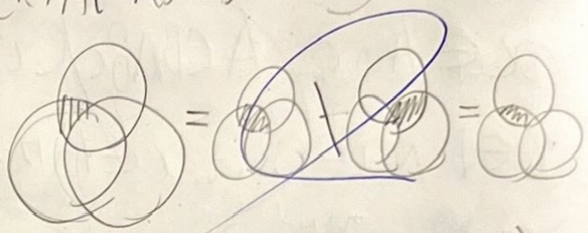


(2) $A\cup B\cup C=U-(\overline{A}\cap\overline{B}\cap\overline{C})$

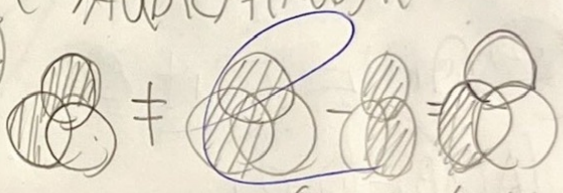
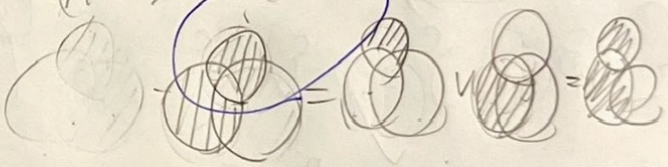


1.7

(1) $A\cap(B\setminus C)=(A\cap B)\setminus(A\cap C)$



(2) $A\cup(B\setminus C)\neq(A\cup B)\setminus(A\cup C)$



(3) $(A\cup B)-(A\cap B)=(A\setminus B)\cup(B\setminus A)$